Semantic Confidence in Embedding-Based Mapping of Patents to Economic Sectors

Giulio Massacci and Mauro Bruno

Abstract Connecting technological innovations to economic sectors is a key challenge when studying how emerging technologies spread. Patent documents provide detailed descriptions of new inventions, but their technological classifications do not directly map to economic activity categories. This paper presents a methodological analysis of embedding-based semantic retrieval for linking patent descriptions to economic sectors. Using a dataset of AI-related patents and NACE sector descriptions, we examine the statistical structure of similarity rankings produced by a multilingual embedding model and introduce a simple semantic confidence measure based on the margin between the top-ranked candidates. Results show that similarity rankings exhibit a consistent structure and that higher confidence values are associated with substantially higher classification accuracy. These findings provide methodological insights into the behaviour of embedding-based classification systems and highlight the potential of simple confidence measures to support more reliable mapping between technological descriptions and economic taxonomies.

Key words: Text embeddings, Semantic similarity, Patent data, Economic classification, Large language models

1 Introduction

The rapid spread of artificial intelligence (AI) technologies is transforming production processes, labor demands, and the structure of economic activities. Understanding how technological innovation impacts different economic sectors is therefore an important research question in economics and official statistics. Patent data serve as a key source of information for examining technological change, as they offer detailed textual descriptions of new inventions. However, linking patents to economic sectors remains difficult because patent classifications reflect technological fields rather than the economic activities in which technologies are applied. Recent research has introduced several methods to evaluate how economic sectors and oc-

Italian National Institute of Statistics (Istat).

cupations are exposed to emerging technologies. For instance, Webb [2020] assesses AI exposure by comparing semantic similarities between patent texts and occupational descriptions.

Despite their growing adoption in empirical research, the statistical behaviour of embedding-based semantic retrieval systems in classification settings remains relatively underexplored (Reimers and Gurevych [2019], Karpukhin et al. [2020]). This paper provides a methodological analysis of embedding-based semantic retrieval for mapping patent descriptions to economic sector classifications. Rather than focusing solely on predictive performance, we investigate the statistical properties of similarity rankings and introduce a simple confidence indicator based on the separation between the top-ranked sectors.

2 Data

The empirical analysis is based on patent abstracts and sector descriptions used in the semantic retrieval framework.

2.1 Patent dataset

Patent data were obtained from Google Patents, a comprehensive open-access platform that aggregates patent documents from major patent offices worldwide. In this study, the patent abstract served as the primary textual input, as it offers a brief overview of the invention and its technological scope. Patents were selected using a set of keywords related to artificial intelligence technologies. Additional filters were applied to limit the dataset to patents filed between 2000 and 2024, written in English. After data extraction and filtering, the final dataset includes 103,554 unique patents.

2.2 Economic sector classification

Economic activities are represented using the NACE Rev. 2.1 classification(Eurostat [2023]). NACE provides a hierarchical taxonomy of economic activities organized into sections, divisions, groups, and classes. For the purposes of this study, we operate at the most detailed level of the classification (four-digit classes), which provides a sufficiently granular representation of economic activities while maintaining interpretability for sector-level analysis.

2.3 LLM-labelled proxy dataset

To evaluate the behaviour of the semantic retrieval system, we constructed a proxy-labeled dataset using a Large Language Model (LLM). For a subset of patents (10% sample), the model was prompted to assign the most relevant NACE class based on the patent title and abstract. The model was instructed to identify the primary economic activity most likely associated with the development, production, or commercialization of the patented invention. The resulting LLM-labeled dataset is used as a proxy for human annotation in the evaluation stage, enabling a consistent comparison between embedding-based sector assignments and model-generated sector labels. While not intended as a ground truth, this proxy provides a coherent and scalable reference for analyzing the statistical behaviour of similarity-based rankings rather than absolute classification accuracy.

3 Methodology

3.1 Embedding-based semantic retrieval

Let p_i denote the textual description of patent i, represented by its abstract, and let s_j denote the textual description of sector j taken from the NACE Rev. 2.1 classification at the four-digit level. Both patent and sector descriptions are mapped into a shared semantic space using a multilingual embedding model. Let

$$\mathbf{e}(p_i),\ \mathbf{e}(s_j) \in \mathbb{R}^d \tag{1}$$

denote the embedding vectors associated with patent p_i and sector description s_j, respectively, where d is the embedding dimension.

Semantic similarity between patents and sector descriptions is measured using cosine similarity between their embedding vectors. For each patent p_i, similarity scores are computed with respect to all sector descriptions, producing a ranked list of candidate sectors:

$$s_{(1)}, s_{(2)}, \ldots, s_{(K)} \tag{2}$$

where $s_{(1)}$ denotes the sector with the highest similarity score and K is the total number of sectors considered.

3.2 Semantic confidence

While the top-ranked sector provides a natural prediction for the patent–sector mapping, the reliability of this assignment may depend on the separation between the first and subsequent candidates. To quantify this aspect, we define a semantic confidence measure based on the margin between the two highest similarity scores:

$$C(p_i) = \mathrm{sim}(p_i, s_{(1)}) - \mathrm{sim}(p_i, s_{(2)}) \tag{3}$$

where $s_{(1)}$ and $s_{(2)}$ denote the first and second ranked sector candidates, respectively. Intuitively, larger values of $C(p_i)$ indicate stronger semantic separation between the top-ranked sector and its closest alternative, suggesting a more reliable classification.

3.3 Evaluation against proxy labels

To evaluate the behaviour of the semantic retrieval system, we compare the retrieved sector rankings with a proxy dataset labelled using a large language model.

For each patent p_i, let y_i denote the set of proxy labels assigned by the LLM, including the primary code and any optional secondary codes. Since this set typically contains one primary sector and only a small number of secondary sectors, the evaluation corresponds to matching the retrieved top-ranked sector against a restricted set of plausible labels. Formally, given a patent p_i:

$$s_{(1)} \in y_i \ \Rightarrow\ \text{correct prediction} \tag{4}$$

We evaluate retrieval performance using standard information retrieval metrics, including Top-1 accuracy and Hit@k (Manning et al. [2008]), and analyse how prediction accuracy varies as a function of the semantic confidence $C(p_i)$.

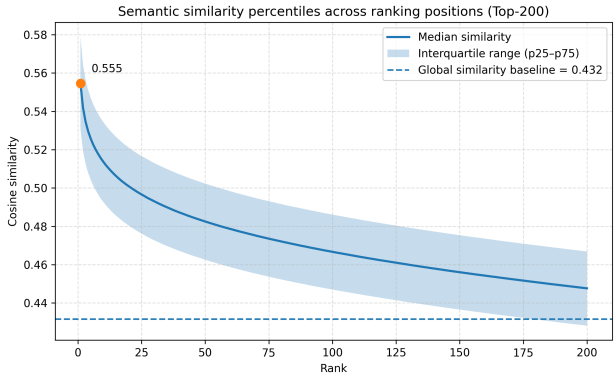

Fig. 1 Median cosine similarity between patent embeddings and NACE sector descriptions as a function of the ranking position. The shaded area represents the interquartile range (p25–p75). The dashed horizontal line indicates the global baseline similarity across all patent–sector pairs.

4 Results

4.1 Structure of semantic similarity rankings

We first analyse the statistical structure of similarity rankings produced by the embedding-based retrieval system (Chen et al. [2024]). For each patent, cosine similarity scores are computed between the patent embedding and each NACE sector description, yielding a ranked list of candidate sectors.

Figure 1 shows the median cosine similarity as a function of the ranking position across patents, together with the interquartile range. The results reveal a clear monotonic decay pattern: similarity values decrease smoothly as the rank increases. The highest-ranked sector typically exhibits a substantially higher similarity score than subsequent candidates, indicating that the semantic retrieval model captures meaningful relationships between patent descriptions and sectoral activities.

The relatively narrow and stable interquartile range across ranking positions suggests that this pattern is consistent across patents, with limited dispersion in similarity scores at each rank.

The dashed horizontal line indicates the overall baseline similarity observed across all patent–sector pairs. The top-ranked sectors consistently appear above this baseline, while lower-ranked candidates gradually move closer to it. This pattern suggests that the embedding space provides a structured representation in which semantically relevant sector descriptions are systematically ranked above unrelated ones.

4.2 Distribution of semantic confidence

We then analyze the distribution of the semantic confidence measure introduced in Section 3. Figure 2 shows the empirical cumulative distribution of the confidence score across patents.

The distribution is strongly concentrated at low values: the median confidence is 0.009 and the mean is 0.013, indicating that for most patents the margin between

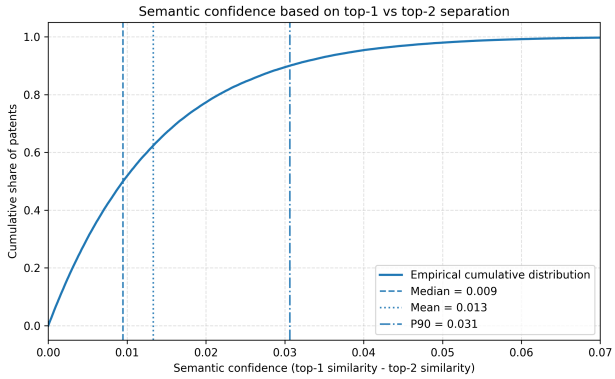

Fig. 2 Empirical cumulative distribution of the semantic confidence score. The vertical lines indicate the median, mean, and 90th percentile of the distribution.

the top two sector candidates is relatively small. The difference between mean and median reflects the right-skewed nature of the distribution, with a relatively small number of high-confidence cases driving the upper tail. This pattern reflects the inherent ambiguity in mapping technological descriptions to economic activities.

At the same time, the upper tail of the distribution highlights a subset of patents with more pronounced separation between candidates. In particular, the 90th percentile is equal to 0.031, meaning that approximately 10% of patents exhibit a confidence level above this threshold, indicating clearer semantic dominance of the top-ranked sector.

Overall, these results indicate that while semantic retrieval often operates in a regime of close competition between candidate sectors, it also produces a structured distribution of confidence values that can be used to distinguish between more and less reliable assignments.

4.3 Confidence and prediction reliability

The overall retrieval performance of the embedding-based system against the LLM-labelled proxy dataset is reported in Table 1. Although Top-1 accuracy remains relatively modest, the progressive increase in Hit@k values indicates that the correct sector is frequently retrieved among the top-ranked candidates. This suggests that the embedding-based retrieval system captures meaningful semantic relationships between patents and economic activities, even when the top prediction is uncertain.

We then analyse the relationship between semantic confidence and classification reliability.

Patents are grouped into quintiles based on their semantic confidence score, from the lowest confidence group (Q1) to the highest (Q5). Top-1 accuracy increases steadily across quintiles, from approximately 0.17 in Q1 to 0.38 in Q5, indicating a strong positive relationship between semantic confidence and prediction accuracy.

Despite the fact that correctness is evaluated against a restricted proxy label set, the observed gradient suggests that semantic confidence provides a useful signal

for identifying more reliable classifications. This supports its potential use in selective validation or human-in-the-loop workflows, where high-confidence cases can be treated differently from more ambiguous ones.

Table 1 Performance of the embedding-based retrieval against the LLM-labelled proxy dataset.

Metric	Value
Top-1 accuracy	0.247
Hit@3 / Hit@5 / Hit@10	0.451 / 0.552 / 0.679
Top-1 accuracy (Q1)	0.17
Top-1 accuracy (Q5)	0.38

5 Conclusion

This paper provides a methodological analysis of embedding-based semantic retrieval for mapping patent descriptions to economic sector classifications, showing that similarity rankings exhibit a consistent structure and that the margin between the top candidates offers a simple and effective signal of prediction reliability. Empirical results indicate that high-confidence cases achieve substantially higher accuracy, while the use of an LLM-based proxy dataset enables scalable evaluation but should be interpreted as an approximate reference rather than ground truth.

These findings highlight the potential of semantic confidence as a practical tool to support more reliable and transparent classification workflows in official statistics. In particular, the proposed measure can be used to identify high-confidence cases for automated processing and low-confidence cases for further validation, enabling more efficient and targeted human-in-the-loop strategies.

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